

1818

7. Write the short notes on any **three** of the following : $3 \times 5 = 15$

- (a) Membranous labyrinth
- (b) Ruminant stomach
- (c) Visceral arches
- (d) Types of feathers

[This question paper contains 4 printed pages]

Your Roll No.

: 30.05.2026(M)

Sl. No. of Q. Paper

: 1818

Unique Paper Code

: 2232012401

Name of the Paper

: DSC- Comparative
Anatomy of Vertebrates

Name of the Course

: B.Sc (H) Zoology
(NEP-UGCF)

Semester

: IV

Time : 2 Hours

Maximum Marks : 60

Instructions for Candidates :

- (a) Write your Roll No. on the top immediately on receipt of this question paper.
- (b) Answer **four** questions in all.
- (c) Question no. 1 is compulsory.
- (d) Draw well-labelled diagrams wherever required.

1. (a) Define any **five** of the following terms :

$1 \times 5 = 5$

- (i) Ungulates
- (ii) Solenoglyphous
- (iii) Brow spot

- (iv) Archinephros
- (v) Craniostyle jaw suspension
- (vi) Merocrine
- (b) Distinguish between any **three** of the following : 2×3=6
 - (i) Larynx and Syrinx
 - (ii) True horns and Antlers
 - (iii) Rod and Cone cells
 - (iv) Single circulation and double circulation
- (c) State exact location and function of the following : 1×4=4
 - (i) Preen gland
 - (ii) Vomeronasal organ
 - (iii) Gizzard
 - (iv) Meibomian gland
- 2. (a) Explain the anatomical details of heart in different vertebrates and draw suitable diagrams. 10
- (b) Define the circulation plan in different mammals. 5
- 3. (a) Discuss the succession of kidney among vertebrates with suitable diagrams. 8

- (b) Describe various types of uteri found in mammals with suitable diagrams. 7
- 4. (a) Describe the various parts of brain. Compare the brain anatomy of reptiles and mammals. 10
- (b) Name the various cranial nerves in mammals with their functions. 5
- 5. (a) Compare the anatomy of digestive tract among amniotes: Draw required diagrams also. 10
- (b) Classify and give functions of various types of receptors found in vertebrates. 5
- 6. (a) Explain the anatomy of avian lung with the help of diagrams and give the mechanism of respiration in birds also. 8
- (b) Describe the structure of integument in vertebrates. Draw appropriate diagrams also. 7